

30. $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ is onto.

- Solution:** By Theorem 10, T has a standard matrix A with 3 rows and 4 columns. This matrix A is unique.
- By Theorem 12(a), the columns of A spans $\mathbb{R}^3$.
 - By Theorem 4, every row of A must have a pivot.
 - From the Row Reduction Algorithm, A can be converted to an echelon form.
 - Let, Row 1 have its pivot in Column 1. By condition 3 from the defn. of echelon form, Row 2 Column 1 = Row 3 Column 1 = 0.
 - Row 1 Column 1 = $\square$
 - By condition 2 from the definition of echelon form, Row 2 will have its pivot in Column 2, 3 or 4.
 - Let Row 2 has its pivot in Column 4. From condition 2 of the defn. of echelon form, ~~Row 2~~ we conclude that A has only 2 pivots. This is a contradiction.
 - $\therefore$ Row 2 must have its pivot in Column 2 or Column 3.
 - Let Row 2 Column 2 = $\square$
 - By condition 3 from the defn. of echelon form, Row 3 Column 2 = 0.
 - By condition 2 from the definition of echelon form, Row 3 will have its pivot in Column 3 or 4.
 - Let Row 3 Column 3 = $\square$
- The final echelon form:
$$\begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & \square & * \end{bmatrix}$$
 * represents those positions which have no constraints.
- Let Row 3 Column 4 = $\square$. By defn of pivot, Row 3 Column 3 = 0.
- The final echelon form:
$$\begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & 0 & \square \end{bmatrix}$$
- Let Row 2 Column 3 = $\square$. By defn. of pivot, Row 2 Column 2 = 0.
 - By condition 3 of the defn. of the echelon form, Row 3 Column 3 = 0.
 - By condition 2 of the defn. of echelon form, Row 3 Column 4 = $\square$